LECTURE 10.1

Ionic Diffusion & Charge Conservation

Battery models with PINNs: from a single particle to electrolyte transport and thermal coupling, on a real cylindrical cell

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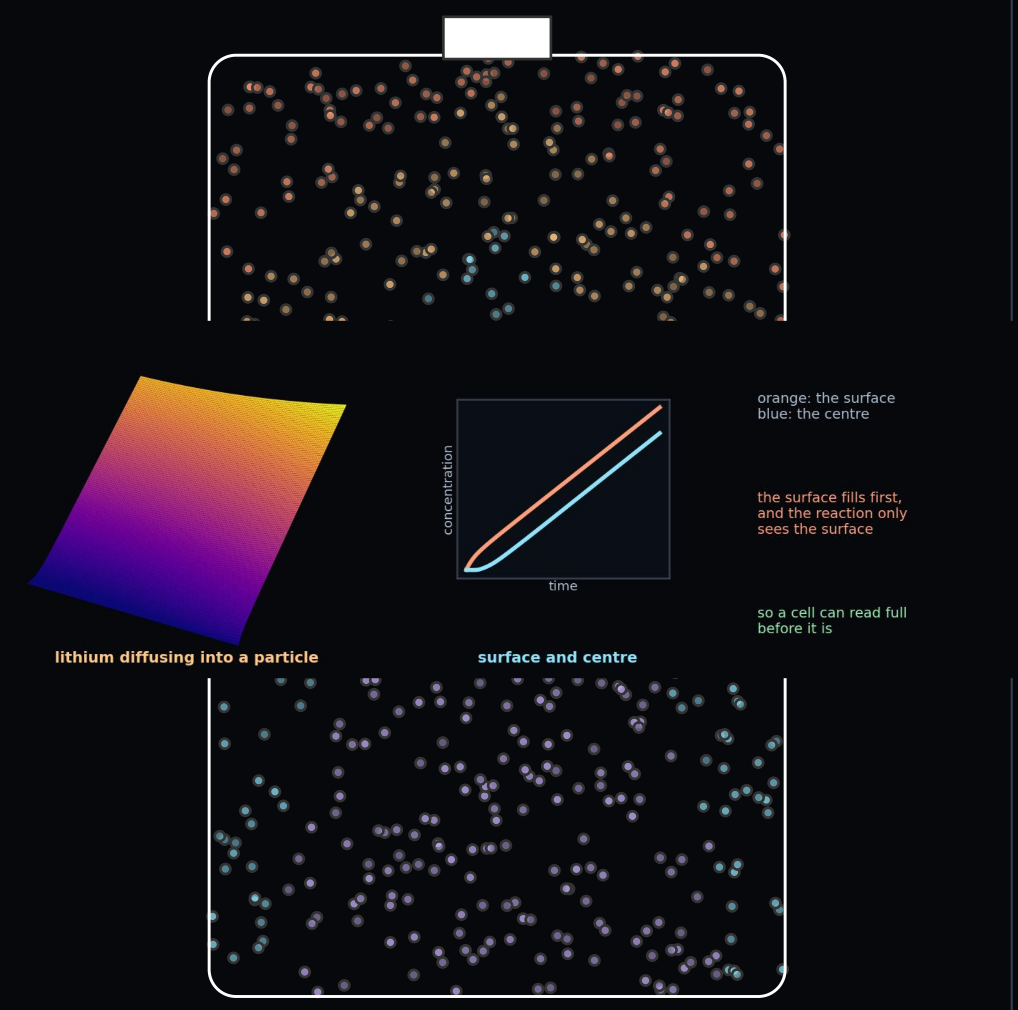
Doyle & Newman DFN · Gu & Wang (2000) · PyBaMM: Chen2020, O'Regan2022 (LG M50 21700)



21700 cell in r–z: hot at the tab, hot at the core

What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **name** — the five fields in a cell model, and the two length scales they live on
- **choose** — between SPM, SPMe and DFN, and defend the choice by C-rate
- **explain** — why the surface concentration matters and the average does not
- **scale** — an electrochemical problem, where raw SI units span twenty orders
- **argue** — why thermal and electrochemical coupling cannot be separated
- **identify** — a diffusivity from a voltage curve, and say when it is not identifiable

WHAT TODAY ASSUMES

from L8.2	the parabolic operator, and time
from L8.1	elliptic charge equations
from L8.2	identifying a parameter from a curve

THE SAME EQUATIONS, NEW NAMES

Solid diffusion is L8.2 in spherical coordinates. Charge conservation is L8.1's Poisson problem. What is new is that they are coupled.

AND ONE SEVERE NONLINEARITY

Butler-Volmer is an exponential. It is the harshest term in the course.

Gu & Wang (2000); PyBaMM docs

The First Genuinely Coupled Multiphysics Problem

	L9 — FLOW	L10 — ELECTROCHEMICAL
Unknowns	*u, v, p*	$c_s, c_e, \varphi_s, \varphi_e, T$
Physics	*one — momentum*	*three — transport, charge, heat*
Coupling	*within one system*	*each physics changes the others*
Nonlinearity	*convection*	*Butler–Volmer, exponential in η*
Dimensions	*x, y, t*	*x, t and a particle radius r*
Scales	*one*	*µm particles, cm cell, hours of discharge*

Every previous lecture solved one physics. This one couples three, and the coupling runs both ways — which is exactly why Gu & Wang argue a decoupled model gets the answer wrong.

Inside the Cell: Five Fields, Two Scales

- **a cell is a sandwich** — collectors, two porous electrodes, a separator
- both electrodes are porous solids flooded with electrolyte
- so lithium lives in two places: inside particles, and dissolved between them
- that gives four electrochemical fields, plus temperature
- **and a second scale** — the particle, where lithium must diffuse to reach the surface

CELL SANDWICH AND THE PARTICLE SCALE

lithium diffuses radially inside each particle

SPM, SPMe, DFN — Choose Deliberately

SINGLE PARTICLE MODEL

SPM

One particle per electrode.
Electrolyte assumed uniform.
Two PDEs, both radial diffusion.
Accurate below about 1C.
Where this lecture starts.

SPM WITH ELECTROLYTE

SPMe

Adds electrolyte concentration and potential along x.
Four fields, still no distributed reaction rate.
Good to roughly 2–3C.

DOYLE–FULLER–NEWMAN

DFN

A particle at every point in x.
Five coupled fields, full Butler–Volmer distribution.
Accurate at high rate.
Research-level for a PINN.

- the hierarchy is an engineering choice, not a ladder of sophistication
- at low C-rate the simplest model is adequate
- and for a PINN the choice matters more than for a classical solver
- each added field is another network, another residual, another thing to balance

Marquis et al. (2019); PyBaMM model hierarchy

Diffusion Inside a Particle

RADIAL DIFFUSION IN A SPHERICAL PARTICLE

$$\frac{\partial c_s}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 D_s \frac{\partial c_s}{\partial r}\right)$$

SYMMETRY AT THE CENTRE, REACTION FLUX AT THE SURFACE

$$\left.\frac{\partial c_s}{\partial r}\right|_{r=0} = 0, \qquad -D_s \left.\frac{\partial c_s}{\partial r}\right|_{r=R_s} = \frac{j}{a_s F}$$

- **this is L8.2's heat equation in spherical coordinates** — the same parabolic operator
- the electrochemistry enters at the surface, through the reaction flux
- and it is the surface concentration, not the average, that the reaction sees
- at high rate the surface saturates long before the centre does
- which is why a cell can appear full while most of the lithium has not arrived

Fick's second law in spherical coordinates

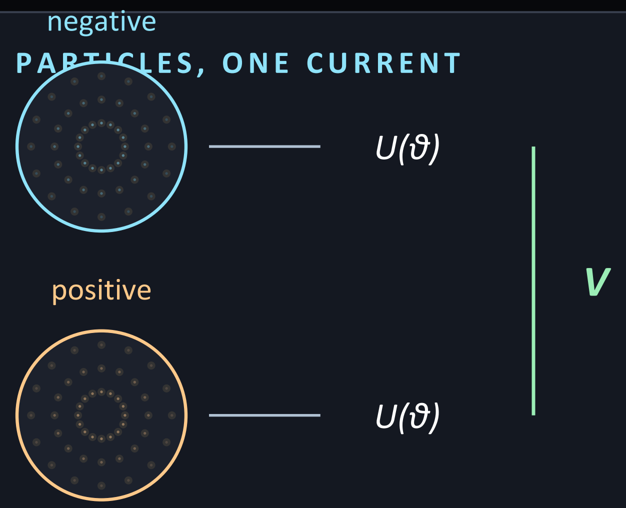
The Single Particle Model

CELL VOLTAGE FROM TWO PARTICLES

$$V = U_p(\theta_p) - U_n(\theta_n) - \eta_p - \eta_n$$

- **one assumption does all the work** — a uniform reaction rate through each electrode
- the model reduces to two radial diffusion problems, coupled only through the current
- the voltage is the difference of two open-circuit potentials, minus the overpotentials
- for a PINN this is an excellent starting point
- two one-dimensional parabolic problems, and one algebraic relation

TWO PARTICLES, ONE CURRENT



no x-dimension: the electrode is one particle

Butler–Volmer: the Reaction at the Interface

CHARGE-TRANSFER KINETICS

$$i_n = i_0 \left[\exp\left(\frac{\alpha_a F \eta}{RT} \right) - \exp\left(-\frac{\alpha_c F \eta}{RT} \right) \right]$$

SURFACE OVERPOTENTIAL

$$\eta = \phi_s - \phi_e - U(\theta, T)$$

EXCHANGE CURRENT DENSITY

$$i_0 = k\, c_e^{\alpha_a} \left(c_{s,\max} - c_{se} \right)^{\alpha_a} c_{se}^{\alpha_c}$$

- two exponentials of the overpotential, one forward and one backward
- this is the most severe nonlinearity in the course
- an exponential amplifies any error in the argument
- **note where temperature enters** — inside the exponential, and again in the potential
- so train on overpotential in units of RT/F, not volts

Gu & Wang Eqs. 3–5; Doyle et al. (1996)

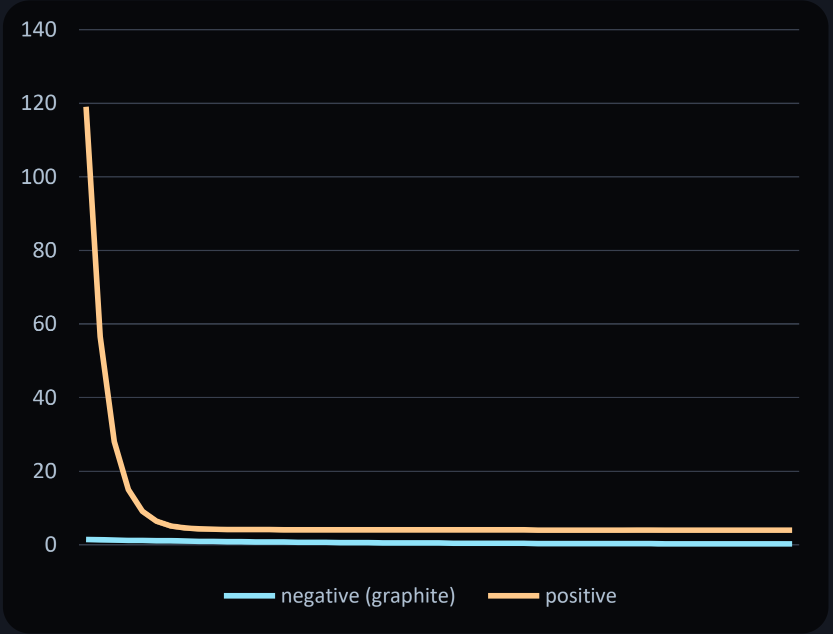
Open-Circuit Potential and State of Charge

LOCAL STATE OF CHARGE

$$\theta = \frac{c_{se}}{c_{s,\max}}$$

- the open-circuit potential is the voltage the electrode would show at rest
- **it is measured, not derived** — supplied as a fitted curve
- strongly nonlinear, with plateaus where a phase transition holds it flat
- the steep ends are where the model is hardest and the cell most vulnerable
- and the temperature dependence is the entropic coefficient

TYPICAL OCP CURVES



Plateaus in the middle, steep walls at both ends. The steep regions set the usable window and are where models break.

Adding the Electrolyte

LITHIUM CONSERVATION IN THE ELECTROLYTE

$$\varepsilon_e \frac{\partial c_e}{\partial t} = \nabla \cdot \left(D_e^{eff} \nabla c_e \right) + \frac{1 - t_+^0}{F} j^{Li}$$

BRUGGEMAN CORRECTION FOR TORTUOSITY

$$D_e^{eff} = D_e \, \varepsilon_e^{1.5}, \qquad \kappa^{eff} = \kappa \, \varepsilon_e^{1.5}$$

- **a parabolic transport equation** — L8.2's heat equation again, with a source
- the source is the reaction: lithium inserted here is lithium removed there
- **the effective diffusivity is not the bulk value** — the path is tortuous
- at high rate the electrolyte depletes at one electrode and accumulates at the other
- that gradient is a real contribution to the voltage loss

Gu & Wang Eq. 10; Marquis et al. (2019)

Two Phases, Two Potentials

CHARGE IN THE ELECTROLYTE — NOTE THE CONCENTRATION-DRIVEN TERM

$$\nabla \cdot \left(\kappa^{eff} \nabla \phi_e\right) + \nabla \cdot \left(\kappa_D^{eff} \nabla ln c_e\right) + j^{Li} = 0$$

CHARGE IN THE SOLID PHASE

$$\nabla \cdot \left(\sigma^{eff} \nabla \phi_s\right) - j^{Li} = 0$$

- **neither charge equation has a time derivative** — both are elliptic
- the Poisson problems of L7.2 and L8.1, in a new setting
- the reaction current appears in both, with opposite signs
- the second term is a diffusion potential, driven by the concentration gradient
- so a full model couples parabolic transport to elliptic charge, at every instant

Gu & Wang Eqs. 17, 20

SUMMARY

Five Equations, Five Unknowns

FIELD	MEANING	TYPE	WHAT IT IS
c_s	lithium in the solid	parabolic, radial	Fick in a sphere
c_e	lithium in the electrolyte	parabolic, along x	with a reaction source
φ_s	solid-phase potential	elliptic	Ohm in the matrix
φ_e	electrolyte potential	elliptic	Ohm plus diffusion potential
T	temperature	parabolic	heat equation with a source

Every one of these five is a PDE you have already solved in isolation. What is new in L10 is solving them together, with each one changing the coefficients of the others.

The Energy Equation and Its Source

ENERGY CONSERVATION — THE L8.2 EQUATION, WITH AN ELECTROCHEMICAL SOURCE

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\lambda \nabla T) + q$$

HEAT GENERATION: REACTION, ENTROPIC, AND OHMIC IN BOTH PHASES

$$q = a_s i_n \eta + a_s i_n T \frac{\partial U}{\partial T} + \sigma_{eff} \nabla \phi_s \cdot \nabla \phi_s + \kappa_{eff} \nabla \phi_e \cdot \nabla \phi_e$$

- three physically distinct sources
- irreversible reaction heat, always positive
- reversible entropic heat, which changes sign with the direction of current
- and ohmic heating in both phases
- **Gu and Wang neglect the entropic term, and say why** — it was not measured then

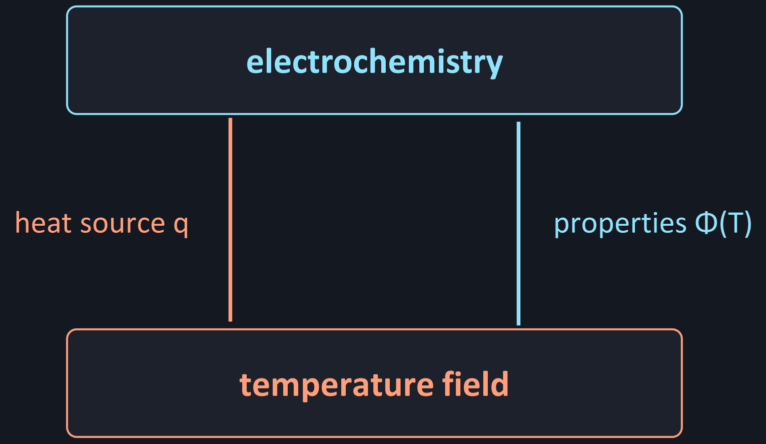
The Loop Runs Both Ways

EVERY PROPERTY IS TEMPERATURE-DEPENDENT

$$\Phi = \Phi_{ref} \exp\left[\frac{E_{act,\Phi}}{R}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right]$$

- electrochemistry generates heat, heat raises temperature, temperature changes transport
- so the loop closes, and Gu and Wang test it directly
- a decoupled model held at reference temperature gets the trend backwards
- the physical reason is that warming a cell helps it
- **so coupling is not a refinement** — decoupling gives qualitatively wrong answers

THE TWO-WAY COUPLING



Drop either arrow and the model is decoupled — and, per Gu & Wang, wrong in both magnitude and trend.

Why the Temperature Field Is Not Uniform

CONVECTIVE COOLING AT THE SURFACE

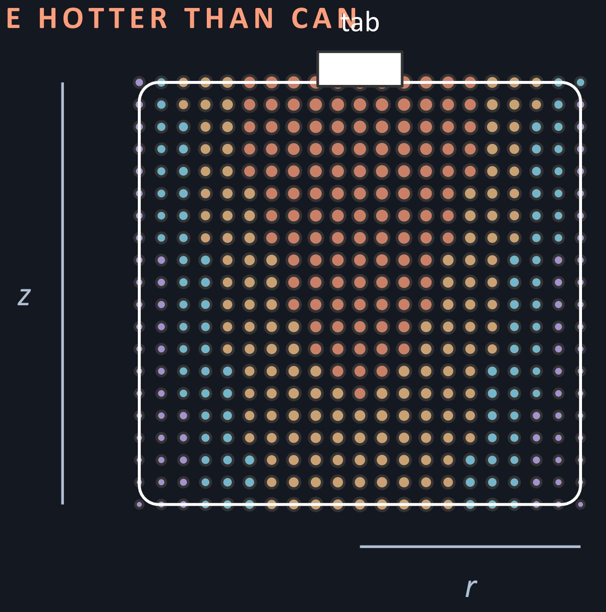
$$-\lambda \frac{\partial T}{\partial n} = h(T - T_a)$$

BIOT NUMBER — IS A LUMPED MODEL ADEQUATE?

$$Bi = \frac{hR}{\lambda}$$

- Gu and Wang report up to 40 °C variation within a single large cell
- a 21700 is smaller, but the mechanism is identical
- heat is generated throughout the jellyroll and leaves only through the surface
- so the natural second dimension is the r–z plane
- **check the Biot number first** — if it is small, a lumped model is defensible

CORE HOTTER THAN CAN



Gu & Wang, 2-D results; cylindrical geometry

PINN FORMULATION

Building the Model, Field by Field

- one network per field, or one network with several outputs
- the radial particle coordinate is a genuine input
- the surface flux condition links the diffusion problem to the reaction
- **and train against a reference** — PyBaMM solves these quickly and reliably
- so the exercise is reproduction, not discovery

WHAT EACH RESIDUAL NEEDS

$c_s(r, t)$	parabolic	$\partial/\partial t$, $\partial^2/\partial r^2$
$c_e(x, t)$	parabolic	$\partial/\partial t$, $\partial^2/\partial x^2$, source
$\varphi_e(x)$	elliptic	$\partial^2/\partial x^2$, $\nabla \ln c_e$
$T(r, z, t)$	parabolic	$\partial/\partial t$, ∇^2, source q
η	algebraic	no derivatives — but exponential

the last row is the hard one: no derivatives, but an exponential

Scaling: Worse Here Than Anywhere Else

- electrochemical quantities span an extraordinary range in SI units
- fed raw, the residuals differ by twenty orders of magnitude between terms
- no weighting recovers from that
- scale concentration by its maximum, radius by the particle radius, time by the discharge
- PyBaMM does this internally, which is the pattern to follow

AN LG M50 IN SI UNITS

QUANTITY	SI	SCALED
particle radius	~5e−6 m	1
solid diffusivity	~1e−14 m²/s	1
$c_{s,max}$	~3e4 mol/m³	1
discharge time	~3600 s	1
overpotential	~0.05 V	$\eta F/RT \sim 2$
capacity	5 Ah	—

Twenty orders of magnitude between the smallest and largest raw quantity. Scaling is not tidiness here — it is the difference between training and not.

PyBaMM as the Reference

- **the LG M50 is the standard public case** — a 5 Ah 21700 cell
- Chen2020 is complete for isothermal work but carries no thermal parameters
- O'Regan2022 measured them, and ships a worked thermal example
- use PyBaMM for the reference solution and the parameters
- and the PINN to reproduce it, then to do what PyBaMM cannot

CHOOSING A PARAMETER SET

Chen2020 — isothermal work
LG M50, original

ORegan2022 — the thermal half
measured thermal data

OKane2022 — ageing studies
superset, degradation

Marquis2019 — not cylindrical
Kokam pouch cell

all four are public and citable

PyBaMM; Chen et al. (2020); O'Regan et al. (2022)

What You Can Recover From a Voltage Curve

DIFFUSIVITY AS A TRAINABLE PARAMETER

$$D_s \; trainable, \qquad \mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_V$$

- a battery management system measures voltage, current and surface temperature
- everything else must be inferred
- that is an inverse problem of exactly the kind L8.2 introduced
- the solid diffusivity ages most and is hardest to measure directly
- **identifiability is the catch** — a relaxation period carries more information than a steady discharge

MEASURED VERSUS INFERRED

MEASURED
$V(t)$, $I(t)$, T_{sur}^{f}

IMPOSED
the model residuals

INFERRED
internal c_s, c_e, T^{core}

INFERRED
D_s, and hence state of health

the core temperature is never measured — and it is the one that matters

L8.2 slide 20; battery state estimation practice

Pitfalls Specific to Electrochemical Models

Unscaled units	Twenty orders of magnitude between raw quantities in one loss.	Symptom: loss barely moves. Fix: non-dimensionalise everything, including η.
Decoupled by accident	Properties left at their reference temperature while the thermal model runs.	Symptom: overpredicted temperature, underpredicted voltage — Gu & Wang, Fig. 2.
Butler–Volmer overflow	An exponential of a large argument during early training produces inf or nan.	Fix: clamp η, or use the linearised form until the model is near a solution.
OCP endpoints	U(θ) is nearly vertical as θ approaches 0 or 1.	Symptom: divergence at end of discharge. Fix: restrict the θ window, or sample it densely.
Wrong model chosen	An SPM used at 3C, where electrolyte depletion dominates.	Symptom: voltage error growing with rate. Fix: SPMe — not more training.

The last is a modelling error, not a numerical one — and it is the one students most often try to fix by training longer.

OVER TO YOU

Questions

TEN TO TEST WHETHER TODAY LANDED

- what are the two length scales, and why does the small one matter?
- why does the reaction see the surface concentration and not the average?
- which equations are parabolic here, and which are elliptic?
- where does temperature enter Butler-Volmer, and why twice?
- why must an electrochemical problem be scaled before training?
- what happens to a decoupled thermal-electrochemical model, and why?

AND FOUR MORE

seven	why is the effective diffusivity not the bulk value?
eight	when is SPM adequate, and when is it not?
nine	what does a BMS measure, and what must be inferred?
ten	PyBaMM already solves this. So why build a PINN?

THE ONE WITH NO SETTLED ANSWER

Which experiment identifies the diffusivity best - and how would you know?

BEFORE L10.2

Ex_10.1 notebooks 01 to 03. Bring your SPM voltage curve.

EXERCISE

Ex_10.1 — From a Single Particle to a Warm Cell

TASKS

1. Generate a PyBaMM reference for the LG M50 at several C-rates, and inspect what changes with rate.
2. Train a PINN for the SPM solid diffusion and reproduce the voltage curve at 0.5C.
3. Extend to SPMe and show where the SPM was wrong — and why that error grows with rate.
4. Add the thermal equation with the Gu & Wang heat source, including the entropic term.
5. Use the panel to vary C-rate, ambient temperature and initial SOC, and report the peak core temperature.

QUESTIONS — AND WHERE THIS DECK ANSWERS THEM

Why does the SPM fail as C-rate rises?	sl. 4, 9
Why must the coupling run both ways?	sl. 13
Which heat term changes sign, and why?	sl. 12
Why is scaling harder here than in L8 or L9?	sl. 16
What can be inferred from a voltage curve alone?	sl. 18

PyBaMM supplies the parameters and the reference. The PINN supplies the differentiable model — which is what L10.2 then optimises.

PERSPECTIVE

PyBaMM Already Solves This — So Why a PINN?

USE PYBAMM

Any forward simulation.
It is faster, validated, and
maintained by the community.
Full DFN with degradation,
ageing and thermal options.
There is no contest here.

USE A PINN

Parameter identification from
measured voltage curves.
Estimating internal states that
no sensor can reach.
A differentiable model that an
optimiser can be run through.

- **say this plainly** — for a forward simulation of a cell, PyBaMM is the better tool
- faster, verified, and maintained by people who do this full time
- the reason to build the PINN is that it is differentiable

The PINN is not competing with the solver. It is producing something the solver does not: a differentiable surrogate constrained by the physics.

L10.2: From a Cell to a Stack

- L10.2 takes the same three physics to a stack
- species transport, charge conservation, and heat
- the equations are recognisably the same
- what changes is the temperature range, and the fact that gradients crack ceramic
- and the channel flow closes the loop with L9

WHAT CARRIES ACROSS

species transport	same operator
charge conservation	same operator
heat with a source	same operator
Butler–Volmer	same form
operating temperature	600–900 °C
gas-phase channel flow	new — from L9

four of six rows are already familiar

Key Takeaways

1 **Five familiar equations**	Parabolic transport, elliptic charge conservation, parabolic heat. Each one you have solved before; what is new is solving them together.
2 **Choose the model, not the biggest**	SPM below 1C, SPMe to about 3C, DFN beyond. Using a DFN where an SPM suffices costs conditioning and buys nothing.
3 **The coupling runs both ways**	Heat changes the properties that generate the heat. Gu & Wang show a decoupled model gets the trend wrong, not merely the magnitude.
4 **Scaling is the first task**	Twenty orders of magnitude between raw quantities. No weighting scheme substitutes for non-dimensionalisation.
5 **The win is what you cannot measure**	Core temperature, internal concentration, state of health. A BMS sees voltage, current and surface temperature — and needs the rest.

Next — L10.2: solid oxide cells, and the real-time optimisation this has all been building towards.

Gu & Wang (2000); Doyle & Newman; PyBaMM

Sources for This Lecture

Gu & Wang (2000)	Thermal-electrochemical coupled modeling of a lithium-ion cell. The source for the thermal coupling, the heat-generation decomposition (Eq. 25), the Arrhenius property dependence (Eq. 26), and the coupled-versus-decoupled comparison.
Doyle, Fuller & Newman	The DFN framework: porous-electrode theory, concentrated-solution transport and Butler–Volmer kinetics. Doyle et al. (1996) supplies the OCP expressions used here.
Chen et al. (2020)	Parametrisation of the LG M50 21700 cell: a complete, public DFN parameter set for a real cylindrical cell. PyBaMM ships it as Chen2020.
O'Regan et al. (2022)	The thermal follow-up to Chen2020, supplying the measured thermal and entropic parameters that Chen2020 lacks. PyBaMM ships it as ORegan2022.
PyBaMM	Open-source battery modelling framework. Used here to generate reference solutions and to supply parameter sets — not as a competitor to the PINN but as the trusted comparison.

Parameter sets and reference data are public and citable — every number in the exercise can be traced to a published measurement.